

Chapter 9 – Linux File System & Directory Structure

Objective

Unlike Windows, Linux organizes files and directories using a hierarchical structure that starts from a single root directory.

Understanding the Linux file system is essential because operating system files, applications, logs, configurations, and user data are stored in different locations.

In this chapter, you will learn:

- Linux directory hierarchy
- Important system directories
- Configuration file locations
- Log locations
- User home directories
- How applications are organized

9.1 Understanding the Linux File System

Linux organizes files in a **tree-like structure**.

Unlike Windows, Linux does **not** use drive letters such as:

C:\

D:\

E:\

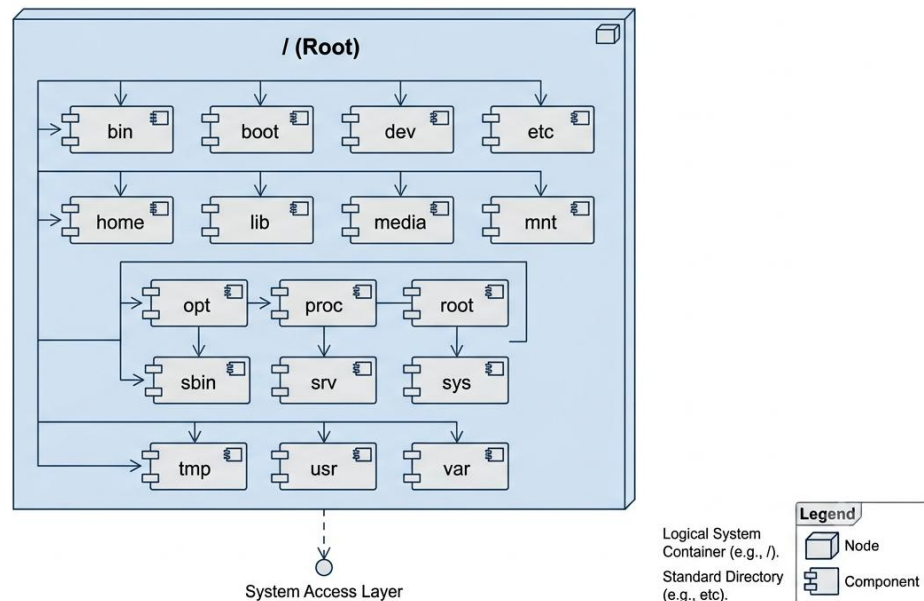
Instead, everything begins from one directory called:

/

This directory is known as the **Root Directory**.

9.2 Linux Directory Hierarchy

UML Deployment Diagram: Linux File System Hierarchy



9.3 The Root Directory (/)

Everything in Linux starts from:

/

The root directory contains all system directories.

It is the highest level of the Linux file system.

Example:

`cd /`

9.4 Important Linux Directories

/home

Stores user home directories.

Example:

`/home/kashifkhan`

This is where personal files are stored.

/root

Home directory of the **root user**.

/root

Normal users should not store files here.

/etc

Stores system configuration files.

Examples:

/etc/hosts

/etc/passwd

/etc/netplan

/etc/ssh

Many server applications store their configuration files here.

Example:

Wazuh:

/etc/wazuh-indexer

/var

Stores files that change frequently.

Examples:

- Logs
- Cache

- Databases
- Mail

Common log location:

/var/log

System administrators frequently check this directory when troubleshooting.

/usr

Contains installed applications and shared libraries.

Most software installed using APT is stored here.

Examples:

/usr/bin

/usr/share

/bin

Contains essential user commands.

Examples:

ls

cp

mv

cat

pwd

These commands are available immediately after boot.

/sbin

Contains administrative commands.

Examples:

shutdown

reboot

fdisk

Most commands require administrative privileges.

/tmp

Temporary directory.

Applications store temporary files here.

Files may be deleted automatically after reboot.

/opt

Used for optional third-party software.

Example:

/opt/google

Some manually installed applications use this directory.

/boot

Contains files required during system startup.

Examples:

- Linux Kernel
- Bootloader

Do not modify these files unless necessary.

/dev

Contains device files.

Examples:

/dev/sda

/dev/null

/dev/random

Linux treats hardware devices as files.

/proc

A virtual filesystem containing information about the running kernel and system processes.

Examples:

/proc/cpuinfo

/proc/meminfo

These files are generated dynamically.

9.5 Viewing the Directory Structure

Display the contents of the current directory.

ls

Display detailed information.

ls -l

Display hidden files.

ls -la

9.6 Display the Current Directory

Show the current working directory.

`pwd`

Example:

`/home/kashifkhan`

9.7 Navigating Directories

Move into a directory.

`cd /etc`

Return to the home directory.

`cd`

Move up one level.

`cd ..`

Go directly to the root directory.

`cd /`

9.8 Why is the File System Important?

Understanding the Linux file system helps administrators locate:

- Configuration files
- System logs
- Installed software
- User data
- Security logs

- Network configuration
- Service files

For example:

Task	Directory
Network Configuration	/etc/netplan
SSH Configuration	/etc/ssh
System Logs	/var/log
User Files	/home
Installed Programs	/usr/bin
Wazuh Configuration	/etc/wazuh-*

Best Practices

- Avoid modifying files inside /boot and /proc.
- Store personal files inside your home directory.
- Keep configuration backups before editing files in /etc.
- Review /var/log when troubleshooting server issues.
- Use absolute paths when performing administrative tasks.

Chapter Summary

In this chapter, you learned:

- The Linux directory hierarchy.
- The purpose of important system directories.
- How to navigate the file system.
- Where Ubuntu stores configuration files, logs, and applications.

- Why understanding the file system is essential for system administration and cybersecurity.
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One-Line Summary

The Linux file system follows a hierarchical structure rooted at /, with each directory serving a specific purpose for storing system files, user data, applications, and configuration information.
